

MambaMPD: A Mamba-Driven Segmentation Framework for Marine Pollution Detection from Remote Sensing Imagery

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ABSTRACT

Accurate detection of marine pollution is essential for protecting coastal ecosystems and marine biodiversity. Recently, vision Mamba-based approaches have shown promise in remote sensing semantic segmentation due to their ability to efficiently capture long-range dependencies and global context. However, their potential remains largely unexplored in the context of Marine Pollution Detection (MPD) with distinct challenges, including low signal-to-noise ratios, spatial fragmentation of pollution patterns, and indistinct boundaries caused by strong visual similarity between pollutants and the surrounding marine environment. To address these challenges, we propose MambaMPD, an enhanced Mamba-based framework tailored for marine pollution detection. **To address these challenges, MambaMPD incorporates two targeted modules that enhance the Mamba encoder with complementary structural priors: the Frequency-Aware Augmentation (FAA) module and the multi-scale Edge-Guided Attention (EGA) module.** The FAA module strengthens the encoding process by integrating Wavelet Transforms, which decompose features into multi-scale frequency subbands. This enables the model to efficiently capture both low-frequency contextual semantics and high-frequency structural details, essential for identifying small, low-contrast, and irregular pollution patterns. Meanwhile, the EGA module adaptively integrates hierarchical Laplacian-derived multi-scale boundary cues into deep semantic representations, guiding the refinement of encoder features before decoding, thereby sharpening boundary delineation and alleviating ambiguity in visually confusing and spatially fragmented marine scenes. Additionally, a U-Net-style decoder equipped with squeeze-and-excitation attention and deep supervision is employed to progressively recover and refine semantic and spatial features across multiple scales. Extensive experiments on two benchmark marine pollution detection datasets demonstrate that **the proposed method achieves competitive or state-of-the-art performance while maintaining significantly lower computational cost compared to foundation-model-based approaches.** On MADDOS, MambaMPD surpasses OSDMamba by 3.6% in F1. On M4D, it achieves a particularly notable 6.82% improvement on the OilSpill category over widely used segmentation architectures, demonstrating strong capability in identifying challenging pollution regions. The source code of our approach will be available at <https://github.com/Multimodal-Intelligence-Lab-MIL/MambaMPD>.

KEYWORDS

Marine pollution detection ; State space model ; Remote Sensing imagery

1. Introduction

Marine pollution monitoring is a highly demanding yet societally vital task. Pollutants such as oil spills [1], plastic debris [2], and chemical contaminants [3] pose serious threats to fragile coastal ecosystems and marine biodiversity, while also disrupting navigation and undermining the sustainability of the blue economy [4–6]. Recent advances in aerospace and sensor technologies have greatly improved access to remote sensing data, including synthetic aperture radar (SAR) [7] and multispectral imagery [8]. Accurately delineating polluted regions from satellite or aerial imagery is therefore crucial for both rapid emergency response and long-term environmental management [9].

Traditional marine pollution detection methods, such as thresholding [10–12], clustering [13, 14], and Markov Random Field-based segmentation [15–17] primarily rely on hand-crafted features like colour, texture, and spatial consistency. While computationally efficient, these approaches often lack robustness and generalisability in complex marine environments. Recent progress has seen a shift toward learning-based models, evolving from CNN-based methods [18–21], to Transformer-based architectures [22–24], and more recently, foundation model-based approaches like SAM-Oil [25]. These developments reflect a transition from task-specific pipelines to prompt-driven, globally-aware segmentation frameworks. More recently, the Mamba architectures [26–28], based on state-space models (SSMs) [26], have gained attention for their ability to efficiently capture long-range dependencies and model global context. While Mamba-based models have shown promise in general remote sensing segmentation [29], their potential remains largely unexplored in the context of marine pollution detection.

Despite significant progress in learning-based methods for marine pollution detection, several fundamental challenges persist across both multispectral and SAR imagery, as illustrated in Fig. 1. These challenges are not specific to any particular architecture but remain largely unresolved across existing CNN-based, Transformer-based, and state-space model-based approaches:

1) Pollution targets such as oil spills and plastic patches typically manifest as small, dispersed regions, often captured by medium-resolution sensors like Sentinel-2 and SAR platforms [7, 30]. These modalities lack the spatial or textural granularity required to capture fine structures, making models highly sensitive to noise [31].

2) Pollutants (such as emulsified oils, rainbow sheens, and plastics) often share similar visual or backscatter patterns with non-pollutant features such as seawater, ship wakes, plankton, and algae [32, 33], leading to substantial semantic ambiguity and misclassification. [34].

3) Dynamic sea surface conditions, driven by waves, currents, and biological activity, further obscure the boundaries between pollution and background regions [35].

While Mamba-based architectures have demonstrated strong capability in capturing long-range dependencies with linear complexity, they still lack dedicated mechanisms to address the above challenges in the context of marine pollution detection. **A recent work, OSDMamba [36], adapts the Mamba architecture for binary oil spill segmentation from SAR imagery but relies solely on the standard VSS encoder without frequency-aware or boundary-guided mechanisms. MambaMPD advances beyond OSDMamba by introducing FAA and EGA modules for improved structural sensitivity, extending the task to multi-class pollution detection, and validating across both SAR and multispectral modalities.** To this end, we propose MambaMPD, a Mamba-driven segmentation framework tailored for marine pollution detection, as shown in Fig. 2 (a). Built upon the expressive capacity of VMamba [27], MambaMPD incorporates several targeted adaptations to address the unique challenges posed by marine pollu-

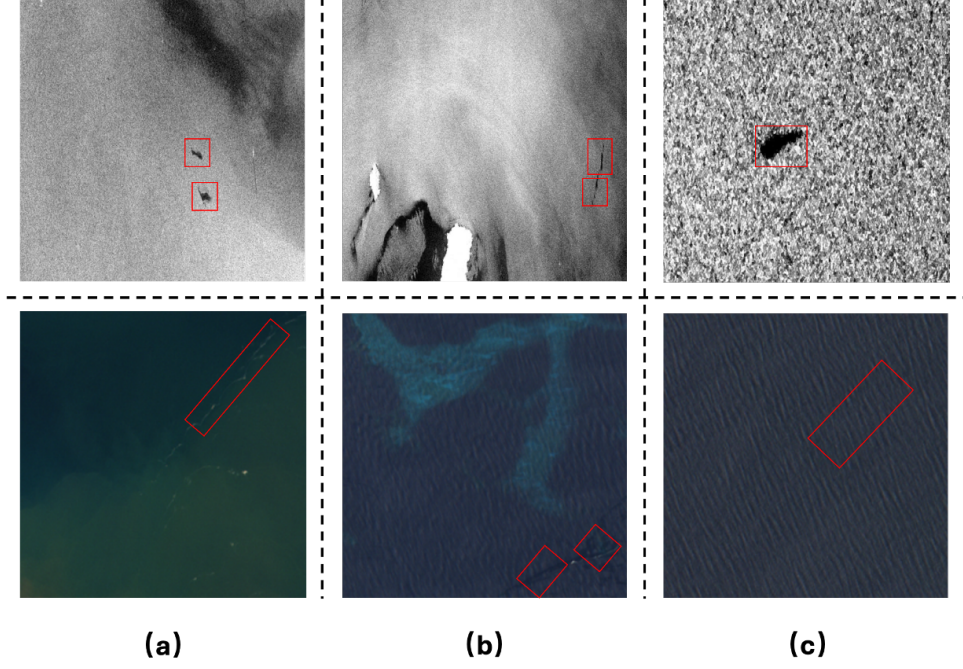


Figure 1. Illustration of key challenges in marine pollution detection across SAR imagery (top row, from M4D dataset) and multispectral imagery (bottom row, from MADOS dataset). (a) Small and dispersed pollution targets (highlighted by red boxes) that are easily obscured by background clutter and noise, making fine-grained detection difficult. (b) Visual and backscatter ambiguity, where oil spills, look-alike dark spots, and natural marine features (e.g., algae, turbid water) exhibit highly similar appearances, leading to frequent misclassification. (c) Indistinct and fragmented pollution boundaries caused by speckle noise in SAR and dynamic sea surface conditions in optical imagery, posing challenges for precise boundary delineation.

tants across both SAR and multispectral imagery. A key component of MambaMPD is the Frequency-Aware Augmentation (FAA) Module, which mainly integrates Wavelet Transforms into the Mamba encoder to decompose features into multi-scale frequency subbands. This allows the model to efficiently capture both contextual semantics and fine structural details, enhancing its ability to detect small, low-contrast, and irregular marine pollution targets.

Another key component of MambaMPD is the multi-scale Edge-Guided Attention (EGA) module, which adaptively integrates Laplacian-based edge cues with semantic features extracted at multiple encoder depths. EGA is specifically designed to enhance boundary preservation under conditions of spatial sparsity. Additionally, a U-Net-style decoder equipped with Squeeze-and-Excitation (SE) attention and deep supervision is employed to progressively recover and refine semantic and spatial features across multiple scales.

To validate the effectiveness of the proposed MambaMPD framework, we conduct comprehensive experiments on two challenging marine pollution detection datasets, M4D [37] and MADOS [30], featuring realistic sea conditions, ambiguous boundaries, and pollutant-background similarities.

In summary, our contributions are shown as follows:

- 1) We propose MambaMPD tailored for marine pollution detection, addressing domain-specific challenges in SAR and multispectral imagery.
- 2) We introduce an FAA module, which augments the Mamba encoder by incorporating WT to capture both low-frequency contextual semantics and high-frequency

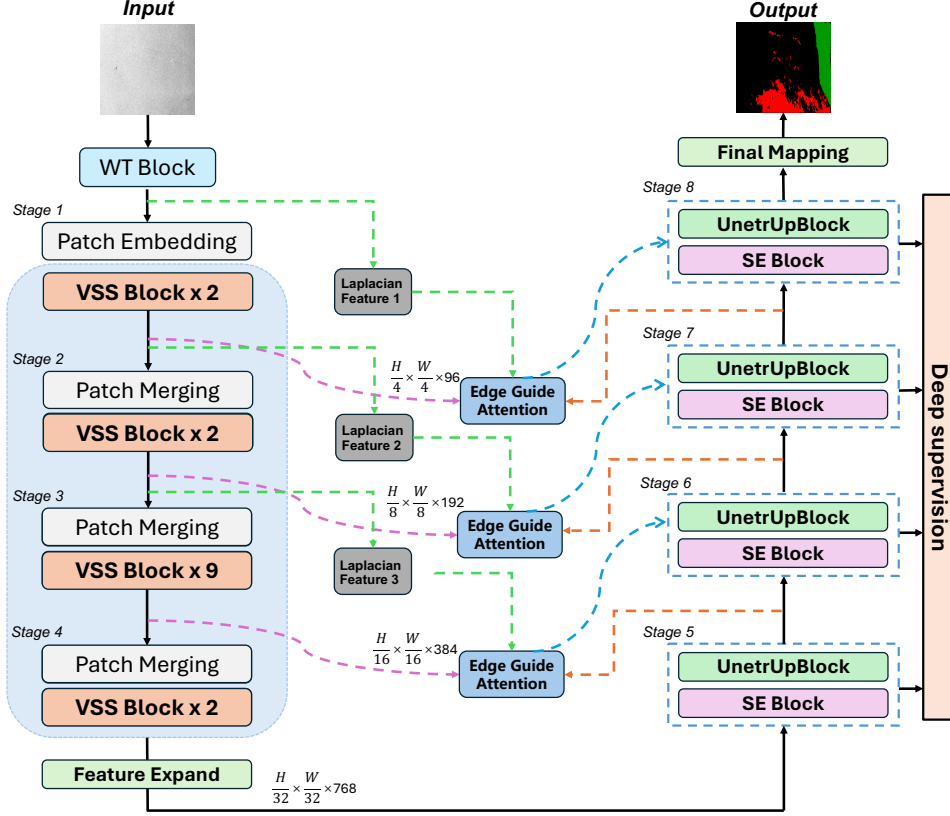


Figure 2. Overall architecture of the proposed framework. The encoder extracts multi-scale features from the input image through a wavelet-enhanced hierarchical backbone consisting of a WT block, patch embedding, patch merging, and stacked VSS blocks. To strengthen boundary modeling, Laplacian features from different stages are incorporated into edge-guided attention modules, which refine encoder representations before they are passed to the decoder. The decoder progressively restores spatial resolution via stacked UnetrUpBlocks and SE blocks, where skip connections explicitly transfer encoder features to the corresponding decoder stages for structural detail recovery. Deep supervision is further applied to intermediate decoding layers, and the final prediction is generated through the output mapping layer.

structural cues, crucial for detecting small, low-contrast, and irregular pollution targets, without introducing noticeable parameter overhead.

3) We introduce a multi-scale EGA module to effectively fuse multi-scale edge features with deep semantic representations. This module improves boundary localisation and reduces confusion in visually similar marine pollution classes.

4) Experimental results show that the proposed MambaMPD achieves state-of-the-art performance, **outperforming TransOilSeg by 6.82% in Oil Spill IoU on M4D and MariNeXt by 4.2% in F1-score on MADOS.**

The framework is modality-agnostic, but the experiments in this paper are limited to oil-spill and marine debris/pollutant segmentation on the M4D (SAR) and MADOS (multispectral) datasets—the two tasks for which public benchmarks currently exist. Other pollution types (e.g. chemical contaminants, microplastics) are discussed as future work in Section 6.

2. Related Work

2.1. Marine Pollution Detection from Remote Sensing

Early approaches to marine pollution detection in remote sensing primarily relied on expert-defined heuristics and rule-based strategies. These traditional methods [32–34, 38, 39], such as spectral thresholding, morphological filtering, and geometric descriptors, were typically tailored to detect a single pollutant type. However, they struggle in multi-class scenarios due to limited adaptability to competing marine elements.

For instance, Kikaki et al. [30] observed that rule-based systems often confuse plastic debris with natural water patterns and misclassify clean water regions when appearance overlap is high. Similar issues have been reported in optical imagery, where marine mucilage is mistaken for floating waste [40], and in SAR data where fine debris is difficult to distinguish from ships or sea clutter due to similar radar backscatter [41]. While fast and interpretable, these methods’ reliance on low-level cues and rigid rules limits their scalability and generalisation in real-world applications.

To address these shortcomings, conventional machine learning (ML) techniques have been explored, where handcrafted features, such as texture, shape, or contextual descriptors, are paired with classifiers like support vector machines or random forests [37, 42, 43]. These approaches introduce greater adaptability and can incorporate spatial patterns to detect complex marine targets. In particular, ML models have shown promise in recognising small-scale or texture-rich pollutants under varying conditions [40]. Nevertheless, their success is often constrained by the quality of manually extracted features and the inherent imbalance in marine datasets. Moreover, specific pollutant types, such as large plastic patches or dispersed oil films, present sporadic appearances and irregular shapes [44, 45], further challenging the generalisation capabilities of these models.

With the rise of deep learning, semantic segmentation techniques have been increasingly adopted in remote sensing applications. Classic encoder–decoder architectures such as U-Net [46] and DeepLab [47] extract hierarchical features and leverage skip connections and atrous convolutions to maintain both global semantic context and fine-grained spatial details [48–50]. These approaches have been progressively extended to the task of marine pollution detection. A series of early works focused on oil spill detection using SAR imagery as a binary segmentation task. For example, Mahmoud et al. [51] employed a standard U-Net to separate oil spills from background water, achieving promising results under simple scenarios. However, this setup lacks scalability for more complex or diverse marine scenes. To address blurry boundaries and noisy textures in SAR imagery, CBD-Net [52] introduced contextual and boundary supervision, enhancing small target segmentation and achieving state-of-the-art performance on the SOS dataset. Further improving SAR-based segmentation, DGNet [53] incorporated the intrinsic distribution of SAR backscatter values into a latent variable inference generation loop, demonstrating data-efficient training and superior generalisation with limited annotations. Similarly, SRCNet [54] designed a competing dual-network structure leveraging a SAR-specific image representation, which enhanced learning from small-scale datasets and achieved accurate segmentation under constrained supervision.

Beyond binary segmentation, recent works began addressing the need to distinguish between multiple types of marine targets in SAR images, such as ship wakes, look-alike dark spots, and other oil-like artefacts. The M4D dataset [7] represents an effort in this direction, offering a richer annotation schema for evaluating fine-grained pollutant

detection. Building on this, Wu et al. [25] proposed SAM-Oil, leveraging vision foundation models for few-shot and interactive segmentation. However, its performance is constrained by prompt quality and the absence of explicit structural cues, limiting precision in noisy, cluttered marine scenes. Chai et al. [21] proposed TransOilSeg, which performs well on large-scale imagery with clear oil slicks but may struggle in scenes with small, fragmented, or low-contrast targets.

To extend beyond SAR, Shen et al. [55] developed a deep learning pipeline for detecting marine debris in multispectral imagery. Kikaki et al. [8] introduced the MADOS dataset and proposed MariNext, a CNN-based model that fuses spectral and spatial cues. While effective across pollutant types and sensors, MariNext lacks mechanisms to capture structural or edge information crucial for complex scenes.

In summary, existing models still struggle with key challenges such as small object size, low contrast and boundary confusion [56–58]. Distinct from prior works, our proposed MambaMPD explicitly addresses these limitations through the introduction of the Frequency-Aware Augmentation (FAA) module and Edge-Guided Attention (EGA) module, which enhance structural sensitivity, preserve contextual semantics, and improve the delineation of fine-grained object boundaries.

2.2. Vision Mamba in Remote Sensing Segmentation

A growing body of research has explored the adaptation of the Mamba architecture [59] to remote sensing image segmentation, showcasing its efficiency in modeling long-range dependencies for high-resolution and large-scale spatial tasks. For instance, RS-Mamba [60] introduces an omnidirectional selective scan to process gigapixel-scale remote sensing images with linear computational cost, achieving competitive results in land cover segmentation and change detection. RS3Mamba [61] complements a convolutional backbone with a state-space auxiliary branch, enabling effective multi-level feature fusion for urban semantic segmentation. In the bi-temporal domain, Change-Mamba [62] employs a Vision-Mamba encoder and a set of specialized decoders to capture temporal correlations, surpassing traditional CNN and Transformer baselines on several change detection benchmarks. Meanwhile, RSMamba [29] further demonstrates the potential of Mamba in spatial classification by introducing dynamic multi-path activation to remove causal constraints and better adapt to 2D spatial tasks. Some preliminary segmentation works, such as in [63, 64], have verified the feasibility of Mamba in land cover mapping and building footprint extraction.

Despite their promising capabilities, existing Mamba-based approaches have not yet been explored in the context of marine pollution detection, a task that requires improved boundary delineation and effective handling of high-frequency structural details, which are essential for detecting small, sparse, and visually ambiguous pollutants. To fill in this gap, our MambaMPD enhances the Vision Mamba architecture with two dedicated modules, FAA and EGA, **introducing an effective adaptation of Mamba tailored for the challenges of marine pollution detection.**

3. METHOD

Marine pollution detection from remote sensing imagery can be formulated as a pixel-wise multiclass semantic segmentation task. Given an input image $X \in \mathbb{R}^{H \times W \times C_{in}}$, the goal is to learn a mapping $Y = \mathcal{F}_\theta(X) \in \mathbb{R}^{C \times H \times W}$ that assigns each pixel a probability distribution over C semantic categories. In our formulation, X can represent

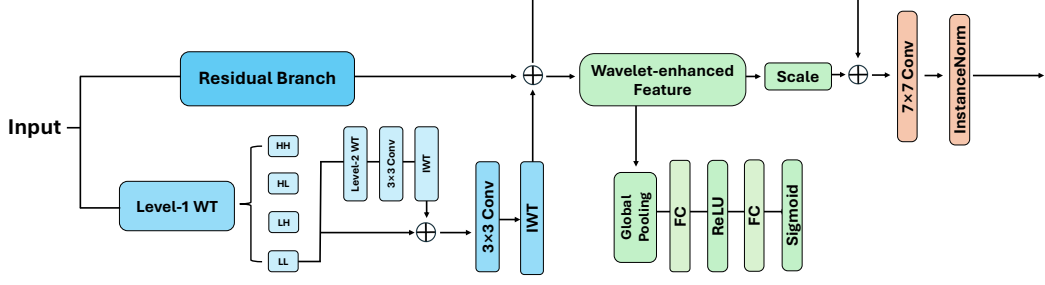


Figure 3. Illustration of the proposed Frequency-Aware Augmentation Module (FAAM). Given an input feature, FAAM employs a dual-branch design consisting of a residual branch and a wavelet transform branch. The wavelet branch performs level-1 wavelet decomposition to separate low- and high-frequency components, while the high-frequency information is further refined through an additional wavelet-based convolution operation. The processed wavelet features are fused with the residual features and subsequently recalibrated by a channel-attention mechanism to emphasize informative frequency responses. The final enhanced representation is produced through residual aggregation, followed by a 7×7 convolution and instance normalization.

either multispectral optical images or single-/dual-polarization SAR data, enabling the framework to operate in a modality-agnostic manner. The following subsections present the proposed MambaMPD framework in detail.

3.1. Mamba-based Encoder

MambaMPD adopts the Visual State Space (VSS) block [65] as its fundamental building unit. The VSS block replaces traditional attention mechanisms with selective state-space dynamics, efficiently modeling long-range dependencies with linear complexity. Given a sequence of visual tokens $x \in \mathbb{R}^{H \times W \times C}$, the core state-space recurrence is defined as:

$$s_t = A \cdot s_{t-1} + B \cdot \text{Conv}_{\text{proj}}(x_t), \quad y_t = \sigma(G_t) \odot (C \cdot s_t), \quad (1)$$

where A and B are learnable transition matrices governing the system dynamics, G_t is a learned spatial gate, and C is a readout matrix. The output is restored to the original spatial dimension via a convolutional projection $\hat{x}_t = \text{Conv}_{\text{out}}(y_t)$. To capture spatial context from multiple orientations, the VSS block further employs a two-dimensional selective scanning (SS2D) mechanism [28] that unfolds the feature map along four directional paths and merges the resulting representations.

Here, $s_t \in \mathbb{R}^N$ denotes the hidden state of dimension N , $x_t \in \mathbb{R}^C$ is the feature vector of a single visual token, and $A \in \mathbb{R}^{N \times N}$, $B \in \mathbb{R}^{N \times C}$, $C \in \mathbb{R}^{C \times N}$ are the state-space matrices. To apply this 1-D recurrence to 2-D feature maps, SS2D [?] unfolds each $H \times W$ feature map along four directional paths (left-to-right, right-to-left, top-to-bottom, bottom-to-top) to produce four sequences of length $H \times W$. Each sequence is processed independently; the four outputs are then summed element-wise and reshaped to $H \times W \times C$. The overall encoder follows the hierarchical design of VMamba-Tiny [65] and Swin-UMamba [28], performing $2 \times$ spatial downsampling at each stage. The four encoder stages are composed of $\{2, 2, 9, 2\}$ VSS blocks, respectively, providing a balanced trade-off between depth and efficiency. However, as shown in Fig. 4(a), our empirical observations reveal that the standard Mamba-based encoder suffers from limited frequency sensitivity and poor edge preservation, particularly for small, sparsely distributed marine pollutants under complex backgrounds.

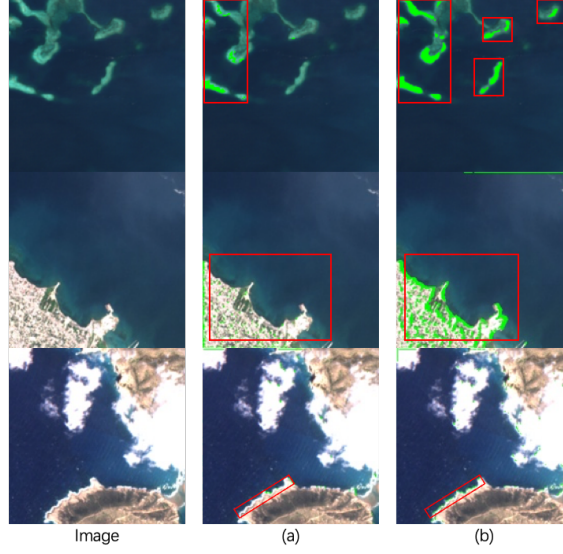


Figure 4. (a) shows the feature response using the baseline Mamba encoder, while (b) demonstrates the enhanced performance achieved by integrating FAA into the baseline Mamba encoder. Red boxes highlight regions where the FAA enhances feature responses in small targets and boundary regions, which are critical for detecting sparse marine pollutants in complex backgrounds.

3.2. Frequency-Aware Augmentation Module

Frequency decomposition is motivated by the physics of marine pollution imagery. In SAR, oil spills dampen capillary and short gravity waves, producing smooth dark patches; the large-scale backscatter contrast is carried by low-frequency components, while the sharp gradient at slick boundaries falls in high-frequency components. In multispectral data, pollutants differ from clean water subtly at a global scale but abruptly at their edges. Wavelet decomposition separates these two cues: low-frequency sub-bands encode region-level semantics and high-frequency sub-bands preserve boundary detail. As shown in Fig. 2, the FAA module sits before the patch embedding layer and operates on the raw input $X \in \mathbb{R}^{H \times W \times C_{in}}$. Placing it this early injects frequency cues before spatial downsampling, so high-frequency detail is not lost to successive patch merging. To address the limited frequency sensitivity of the standard encoder, we propose a Frequency-Aware Augmentation (FAA) module, as illustrated in Fig. 3. FAA introduces hierarchical wavelet decomposition together with lightweight convolutional refinement, enabling the encoder to simultaneously capture global contextual structures and fine-grained boundary details that are critical for marine pollution segmentation.

Given an input feature tensor $X \in \mathbb{R}^{C \times H \times W}$, we first apply a separable 2D Haar wavelet transform to decompose X into four frequency sub-bands:

$$[X_{LL}, X_{LH}, X_{HL}, X_{HH}] = \text{WT}(X), \quad (2)$$

where X_{LL} denotes the low-frequency approximation component, and X_{LH} , X_{HL} , X_{HH} capture high-frequency details along the horizontal, vertical, and diagonal directions, respectively. Each sub-band is refined by a shared 3×3 convolutional operator $\phi(\cdot)$ and reconstructed via the inverse wavelet transform (IWT).

To further enlarge the effective receptive field and strengthen multi-scale fre-

quency perception, we adopt a hierarchical decomposition strategy where only the low-frequency approximation branch from the previous level is further decomposed:

$$[X_{LL}^{(i)}, X_{LH}^{(i)}, X_{HL}^{(i)}, X_{HH}^{(i)}] = \text{WT}(X_{LL}^{(i-1)}). \quad (3)$$

This enables the model to progressively capture increasingly global structures while preserving high-frequency details at each level. The features are then reconstructed in a coarse-to-fine manner:

$$Z^{(i)} = \text{IWT}(y_{LL}^{(i)} + \uparrow(Z^{(i+1)}), y_{LH}^{(i)}, y_{HL}^{(i)}, y_{HH}^{(i)}), \quad (4)$$

where $y_{LL}^{(i)}$, $y_{LH}^{(i)}$, $y_{HL}^{(i)}$, and $y_{HH}^{(i)}$ are the refined sub-band features at level i , $\uparrow(\cdot)$ denotes upsampling to match the spatial resolution of the current level, and $Z^{(i+1)}$ is the reconstructed output from the next coarser level. Through this top-down reconstruction scheme, FAA progressively integrates coarse contextual information with fine structural details across decomposition levels.

The DWT and IWT use **fixed Haar wavelet filters** implemented as stride-2 convolutions and transposed convolutions, respectively; their weights are frozen and receive no gradient updates. The only trainable components in FAA are the 3×3 refinement convolutions $\phi(\cdot)$ applied to each sub-band and the subsequent SE channel attention block. Gradients pass through the fixed wavelet layers via the standard chain rule, as through any non-learnable linear operation. Table 10 confirms this: **FAA adds no trainable parameters**. The reconstructed feature is then fused with a residual branch and passed through a squeeze-and-excitation (SE) block to adaptively recalibrate channel responses. Finally, a 7×7 depthwise convolution followed by instance normalization is applied to enhance local-global interaction and generate the final augmented feature. As illustrated in Fig. 4(b), incorporating FAA into the encoder leads to more discriminative responses in small-target and boundary regions. **The ablation study (Table 5) confirms this quantitatively: FAA alone improves Oil Spill IoU by +5.29% over the baseline**, which are particularly important for detecting sparse marine pollutants under complex backgrounds. Notably, as reported in Table 10, this additional cost is negligible in practice, increasing FLOPs by only 0.03G and inference latency by less than 0.1 ms.

3.3. Decoder

To improve feature recalibration and semantic refinement during decoding, we adopt a Squeeze-and-Excitation Residual Decoder (SE-ResDecoder), as shown in Fig. 2. Each decoder stage consists of a UnetrUpBlock followed by an SE block. The UnetrUpBlock first upsamples the coarser decoder feature via transpose convolution and concatenates it with the corresponding encoder skip feature. The fused representation is then refined through two residual convolutional blocks composed of convolution, normalization, and LeakyReLU activation. This design enables the decoder to progressively align semantic information from coarse decoder features with structural details from shallow encoder features. Subsequently, a squeeze-and-excitation block performs channel-wise recalibration to emphasize the most informative feature channels.

In addition, deep supervision is applied at each decoder stage: a 1×1 convolution generates an auxiliary segmentation prediction that is upsampled to the original resolution and supervised by the ground-truth mask, facilitating stable gradient propagation

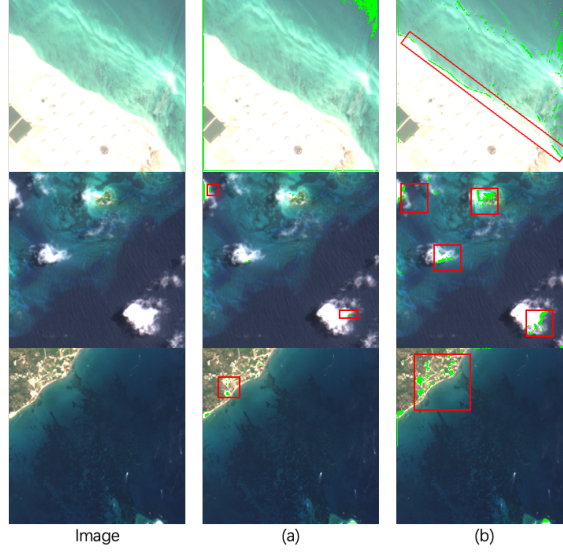


Figure 5. (a) presents the baseline feature response, and (b) shows the enhanced feature response with the SE-ResDecoder. Red boxes show that our approach produces more accurate boundaries and consistent segmentation masks, demonstrating improved robustness in small-scale and ambiguous marine pollutant regions.

and encouraging consistent semantic recovery across decoder stages. As illustrated in Fig. 5(b), the proposed SE-ResDecoder enables the network to recover more semantically coherent and spatially precise predictions, particularly in challenging regions with ambiguous boundaries, fragmented structures, and small targets.

3.4. Edge-Guided Attention (EGA) Module

Although the Mamba-based encoder–decoder architecture exhibits strong global contextual modeling capabilities, Fig. 6(a) shows that purely semantic decoding often struggles to recover fine-scale pollution structures, particularly in low-contrast or visually ambiguous ocean regions. These failure cases suggest that decoder features alone are insufficient to provide the geometric cues required for accurate boundary reconstruction. To address this limitation, we introduce an Edge-Guided Attention (EGA) module, as schematically illustrated in Fig. 7. The purpose of EGA is to inject structural guidance into the decoding process by jointly exploiting global context, local structural cues, and edge-aware guidance derived from the decoder prediction.

Let $\hat{f}_i^e \in \mathbb{R}^{H_i \times W_i \times C_i}$ denote the encoder feature at stage i , and let $\hat{f}_{i+1}^d \in \mathbb{R}^{H_i \times W_i \times 1}$ denote the decoder-side prediction aligned to the same spatial resolution. $\hat{f}_{i+1}^d$ is *not the final segmentation mask but an intermediate auxiliary prediction from the deep supervision branch at decoder stage $i+1$* . As described in Section 3.3, each decoder stage produces a single-channel prediction via a 1×1 convolution and sigmoid activation. This prediction is bilinearly interpolated to the resolution of the encoder feature $\hat{f}_i^e$ before entering EGA. Using these stage-wise predictions as edge guidance lets EGA access semantic cues that sharpen as decoding progresses, rather than depending on a *single fixed output*. Following the upper-left branch of Fig. 7, we first derive complementary global and local spatial response maps f_i^g and f_i^{loc} from the encoder feature via a Global Feature Extractor and a Local Feature Extractor, respectively, where f_i^g captures broad contextual saliency and f_i^{loc} preserves local structural details. To ex-

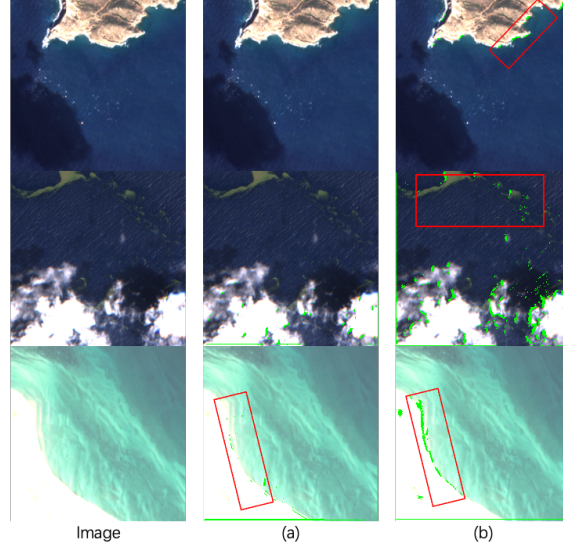


Figure 6. (a) represents the baseline model without edge guidance and (b) represents the baseline model with the proposed EGA module. Red boxes highlight that the baseline with the EGA module produces sharper boundaries and more coherent feature responses.

plicitly inject boundary cues, the decoder prediction is transformed into an edge-aware map through a Laplacian operator:

$$f_i^{edge} = \left| \mathcal{L} * \hat{f}_{i+1}^d \right|, \mathbf{L} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad (5)$$

The global and local responses are combined and modulated by the edge prior to form a structurally guided attention map:

$$f_i^m = (f_i^g + f_i^{loc}) \odot f_i^{edge}, \quad (6)$$

where $\odot$ denotes element-wise multiplication. The resulting map encodes structurally guided spatial attention informed by both context and predicted boundaries.

This modulated map is concatenated with the encoder feature, refined by a 3×3 convolution, and combined with the original encoder feature through a residual connection to preserve the original representation while injecting guided spatial emphasis:

$$f_i^a = \text{Conv}_{3 \times 3}(\text{Concat}[\hat{f}_i^e, f_i^m]) + \hat{f}_i^e. \quad (7)$$

Finally, an Efficient Channel Attention (ECA) block [66] is applied to adaptively emphasize informative channels. The ECA block computes a channel descriptor via global average pooling and obtains attention weights through a lightweight 1D convolution, producing the edge-enhanced output f_i^{out} for subsequent decoding. A detailed algorithmic summary of the full EGA pipeline, including multi-scale edge prior construction and cross-scale feature fusion, is provided in Table 1. As shown in Fig. 6(b), the proposed EGA module substantially improves boundary fidelity, reduces false positives in ambiguous regions, and facilitates coherent segmentation of thin, irregular, and highly fragmented pollution structures.

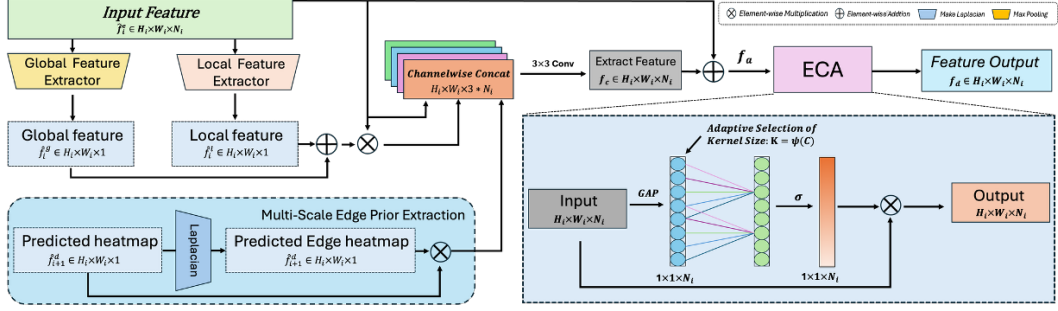


Figure 7. Conceptual overview of the proposed Edge-Guided Attention (EGA) module. The encoder feature is processed by parallel global and local feature extractors to capture complementary contextual and detailed spatial cues. In parallel, the decoder prediction is transformed into an edge-aware guidance map, which is used to modulate the feature representation. The modulated features are then concatenated and refined, followed by channel-wise recalibration using the Efficient Channel Attention (ECA) module, to produce the final edge-enhanced output. This figure is intended as a schematic illustration of the overall interaction mechanism, while the detailed implementation and mathematical formulation are provided in Eqs. (5)-(7).

Table 1. Pseudo-code of the proposed Edge-Guided Attention (EGA) at decoder stage i .

Input: Encoder feature $f_i^e \in \mathbb{R}^{C_i \times H_i \times W_i}$, decoder prediction f_{i+1}^d
Output: Refined decoder feature f_i^d
Phase I: Multi-scale edge prior construction
1. Fine-scale Laplacian response: $E^{(1)} \leftarrow \mathcal{L} * f_{i+1}^d $.
2. For each coarse scale $k \in \{2, 4\}$, compute $\tilde{f}_{i+1}^{(k)} \leftarrow \text{AvgPool}(f_{i+1}^d; k)$ and $E^{(k)} \leftarrow \text{Upsample}(\mathcal{L} * \tilde{f}_{i+1}^{(k)})$.
3. Aggregate cross-scale edge cues: $f_i^l \leftarrow \text{Norm}(\frac{1}{3} \sum_{k \in \{1, 2, 4\}} E^{(k)})$.
Phase II: Edge-aware channel modulation
4. Global semantic descriptor: $z_{\text{glob}} \leftarrow \text{GAP}(f_i^e)$.
5. Edge-aware channel descriptor: $z_{\text{edge}}(c) \leftarrow \frac{\sum_p f_i^l(p) f_i^e(c, p)}{\sum_p f_i^l(p) + \varepsilon}$, $c = 1, \dots, C_i$.
6. Descriptor fusion: $z \leftarrow \lambda z_{\text{glob}} + (1 - \lambda) z_{\text{edge}}$.
7. Channel reweighting: $w \leftarrow \sigma(\text{Conv1D}(z))$, $\hat{f}_i^e \leftarrow f_i^e \otimes w$.
Phase III: Cross-scale feature fusion and decoder refinement
8. Feature concatenation: $f_i^c \leftarrow \text{Concat}(\hat{f}_i^e, f_i^l, f_{i+1}^d)$.
9. Decoder refinement: $f_i^d \leftarrow \text{Conv}_{1 \times 1}(f_i^c)$.

4. Experiments

4.1. Datasets

1) **MADOS Dataset** [30]: The MADOS (Marine Debris and Oil Spill) dataset is a globally distributed benchmark dataset specifically designed for detecting marine pollution, including oil spills and marine debris. It contains 174 multispectral Sentinel-2 (S2) satellite images, captured between 2015 and 2022, with approximately 1.5 million pixels annotated. The dataset covers 15 different thematic categories, such as oil spills, marine debris, ships, phytoplankton, turbid waters, etc. The S2 images in the MADOS dataset cover global coastal waters with spatial resolutions of 10 meters, 20 meters, and 60 meters, and a revisit cycle of about 5 days. During the data processing, ACOLITE atmospheric correction [80] was used to extract Rayleigh reflectance, and annotations were provided by multiple experts, with a focus on labelling major pollutants like oil spills and marine debris. Additionally, the dataset exhibits significant diversity, covering various geographic distributions and environmental conditions, capturing different weather and ocean states. **We adopt the official scene-level data split provided**

Table 2. Quantitative comparison of MambaMPD on the MADOS dataset(Excluding foundation model).

Model	F1 (% , $\uparrow$)	mIoU (% , $\uparrow$)	OA (% , $\uparrow$)
RF[67]	56.6	43.9	67.1
RF++[68]	64.4	52.4	83.8
U-Net[46]	63.8	51.0	82.9
SegNext[69]	60.6	49.2	86.6
MariNeXt [30]	70.6	59.2	81.6
OSDMamba[36]	71.2	68.1	82.3
MambaMPD (ours)	74.8	69.8	83.1

Table 3. Quantitative comparison of our MambaMPD model against recent remote sensing foundation models on the MADOS dataset.

Model	mIoU (% , $\uparrow$)
DOFA[70]	59.58
GFM-Swin[71]	64.7
prithvi[72]	49.9
RemoteCLIP[73]	60.0
SatlasNet[74]	55.9
Scale-MAE[75]	57.3
SpectralGPT[76]	57.9
SSL4EO-S12-MoCo[77]	51.8
SSL4EO-S12-DINO[77]	49.4
SSL4EO-S12-MAE[77]	49.9
SSL4EO-S12-Data2Vec[77]	44.4
TerraMind-B (TerraMesh)[78]	69.5
PANGAEA(U-Net)[79]	57.8
PANGAEA(VITB-16)[79]	48.2
MambaMPD (ours)	69.8

by Kikaki et al. [30], where training and test samples correspond to geographically distinct Sentinel-2 acquisitions, preventing spatial leakage between partitions.

2) **M4D Dataset** [7]: The M4D dataset was developed for marine oil spill detection research. The dataset consists of images extracted from satellite synthetic aperture radar (SAR) data, covering oil spills and other related semantic categories, with corresponding ground truth masks and labels [7]. Specifically, the dataset comprises 1002 training images and 110 test images, each with corresponding labels. A total of 5 semantic classes are annotated, including: sea surface, oil spill, look-alike, ship, and land. We use the fixed 1002/110 train/test split of Krestenitis et al. [7]; training and test images come from separate SAR acquisitions with no spatial overlap. We hold out 5% of the training set for validation.

4.2. Implementation Details

Our network is built using the PyTorch framework. In the M4D dataset, the training template we used follows the design in HTSM [81]. We trained the model using

Table 4. Quantitative performance comparison on the M4D dataset.

Model	Sea Surface(%)	Oil Spill(%)	Look alike(%)	Ship(%)	Land(%)	mIoU(%)
Unet[83]	93.90	53.79	39.55	44.93	92.68	64.97
LinkNet[84]	94.99	51.53	43.24	40.23	93.97	64.79
PSPNet[85]	92.78	40.10	33.79	24.42	86.90	55.60
DeepLabv2[86]	94.09	25.57	40.30	11.41	74.99	49.27
DeepLabv2 (msc)[86]	95.39	49.28	31.26	88.65	93.97	62.83
DeepLabv3+[47]	96.43	53.38	55.40	27.63	92.44	65.06
YOLOv8-SAM[25]	94.34	41.84	48.15	52.48	87.65	64.89
SAM-OIL[25]	96.05	51.60	55.60	52.55	91.81	69.52
TransOilSeg[21]	97.02	61.38	62.41	33.49	94.39	69.74
OSDMamba[36]	96.47	65.59	47.57	46.85	94.76	70.25
MambaMPD (ours)	96.30	68.20	43.60	49.39	93.79	70.85

an SGD optimizer with a momentum of 0.9 and a weight decay of $1e-4$. Additionally, we set the initial learning rate to 0.001 and employed a “Poly” decay strategy. All experiments were implemented on an NVIDIA L40 48GB GPU. The batch size was set to 8, with a maximum of 100 epochs. Similarly, in the MADOS dataset, we used the official training framework provided by [8]. For classical CNN methods, the backbone networks employed during the encoder stage were pretrained on ImageNet. Following OSDMamba [36], we use a hybrid objective that combines Focal Loss and Jaccard Loss, formulated as $L = -\alpha(1 - p_t)^\gamma \log(p_t) + \left(1 - \frac{\sum_i TP_i}{\sum_i TP_i + \sum_i FP_i + \sum_i FN_i}\right)$, where α balances class frequencies, p_t is the predicted probability of the true class, γ modulates the emphasis on hard examples, and TP_i, FP_i, FN_i denote pixel-wise statistics for class i . The model was initialized with pre-trained weights from ImageNet [82]. A batch size of 4 was used, and the model was trained for 100 epochs across all stages. Deep supervision is applied across decoding stages, and the final training objective sums the hybrid loss with all auxiliary prediction losses using pre-defined layer-wise weights. **All baseline results in Tables 2–4 are taken from the original publications. Foundation-model results on MADOS come from the PANGAEA benchmark [79], which uses the same fine-tuning and evaluation protocol for all compared models. M4D provides single-channel SAR intensities replicated to three channels and normalised with mean=0.5185 and std=0.197. Because the input is already three-channel, ImageNet-pretrained weights load directly into the patch embedding layer without modification. For MADOS, we follow Kikaki et al. [8]: all 11 Sentinel-2 bands are resampled to 10 m resolution by nearest-neighbour interpolation and normalised with per-band statistics. The patch embedding layer is re-initialised with $C_{in} = 11$ (random weights) to handle the 11-channel input; all other encoder layers keep their ImageNet-pretrained weights.**

4.3. Evaluation Metrics

For the M4D dataset, we use the mean Intersection over Union (mIoU) score over the union to evaluate the model performance. In the MADOS dataset, both mIoU and average F1 (Ave.F1) scores are used to evaluate the model performance. These two evaluation metrics are based on the confusion matrix, which contains four components: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). For each class, IoU is defined as the ratio of the intersection and union of the predicted and ground truth values. **In addition, we report macro-averaged Precision, Recall, and F1-score on the M4D dataset (Table 6) to provide a more comprehensive assessment of model performance.**

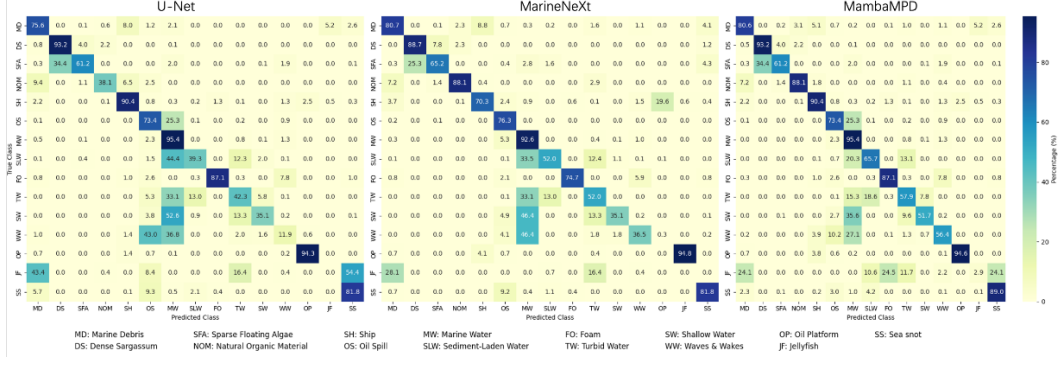


Figure 8. Comparison of confusion matrices on the MADOS dataset.

4.4. Comparison With State-of-the-Art

4.4.1. Quantitative Results on MADOS Dataset

We evaluated and compared our MambaMPD with RF [67], RF++ [68], U-Net [46], SegNeXt [69], MariNext (reproduced) [30], and approaches involving fine-tuning of remote sensing foundation models, such as CROMA, DOFA, GFM-Swin, Prithvi, RemoteCLIP, SatlasNet, Scale-MAE, SpectralGPT, SSL4EO, TerraMind-B, and PAN-GAEA [70–79, 87]. Table 2 and Table 3 report the segmentation performance of each method on the MADOS dataset, further demonstrating the effectiveness of our proposed MambaMPD. Our MambaMPD achieves an F1 score of 74.8% and an mIoU of 69.8%, outperforming the other compared methods. Compared to classical ensemble methods such as RF (F1: 56.6%, mIoU: 43.9%) and RF++ (F1: 64.4%, mIoU: 52.4%), MambaMPD shows substantial improvements, particularly in terms of segmentation quality. When compared to U-Net [46] and SegNeXt [69], which are popular CNN-based architectures, MambaMPD delivers significant gains of 5.5% and 20.1% in mIoU, respectively. Moreover, it also outperforms MariNext [30], a multi-branch CNN designed for multi-type marine pollutant recognition, by a notable margin of 10.1% in mIoU and 4.2% in F1 score. These results clearly demonstrate the superior capability of MambaMPD in handling complex and heterogeneous marine scenes. Its strong performance across all three metrics reflects not only accurate segmentation but also robustness in classifying fine-grained and ambiguous targets under real-world oceanic conditions.

Furthermore, we computed the confusion matrix obtained by applying MambaMPD on the MADOS test set, as shown in Fig. 8. Our model shows better performance compared with other models. Although the number of test pixels for oil spill categories is relatively low compared to others, our model still achieves good segmentation performance. Specifically, the F1 scores for small-object classes consistently exceed 70%, reflecting the model’s robustness in addressing class imbalance. Notably, the accuracy for the Ship class reaches 90.4%, indicating that our model exhibits strong discriminative capability for key categories, effectively reducing misclassification.

4.4.2. Quantitative Results on M4D Dataset

We further evaluated and compared the proposed MambaMPD with several existing methods, including Unet [46], LinkNet [84], PSPNet [85], Deeplabv2 [86], Deeplabv2 (msc) [86], Deeplabv3+ [47], YOLOv8-SAM [25], SAM-OIL [25] and TransOilSeg [21]

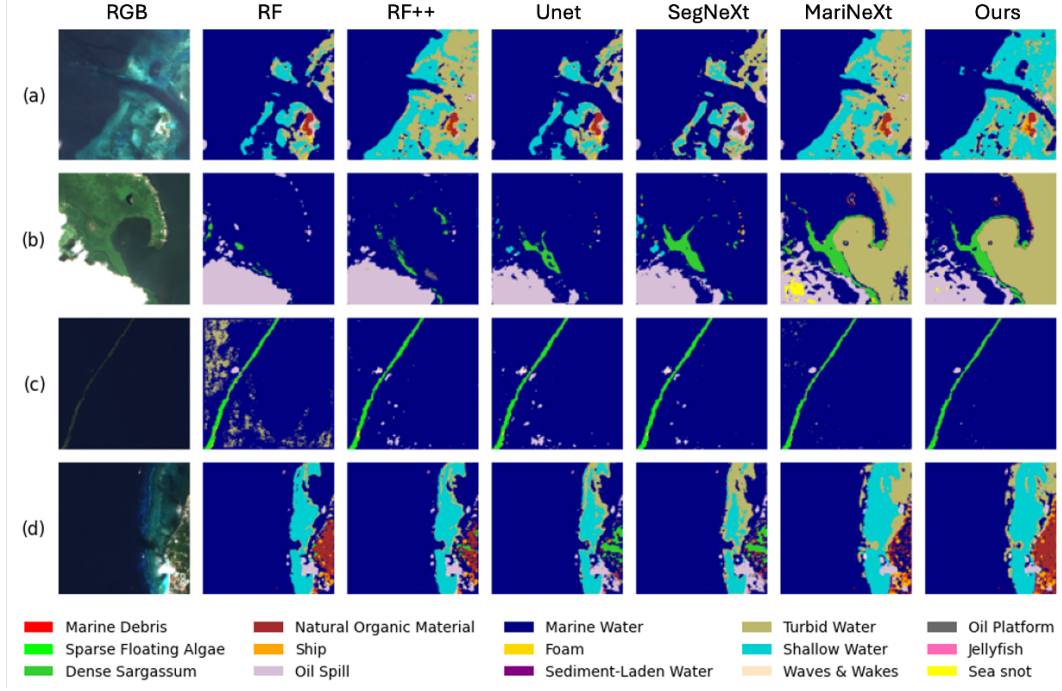


Figure 9. Qualitative performance comparison on selected test samples of the MADOS dataset.

on the M4D dataset.

Table 4 demonstrates that our MambaMPD outperforms other methods in both oil spill mIoU and total mIoU. We further compared MambaMPD with models built upon the Segment Anything Model (SAM), specifically YOLO-SAM and SAM-OIL, which leverage prompt-based or detection-driven strategies to address oil spill segmentation. While these methods benefit from SAM’s general-purpose segmentation capability, they lack dedicated mechanisms for modelling structured context or preserving fine-grained boundaries. In contrast, our MambaMPD leverages the selective scanning mechanism of the Mamba architecture to explicitly capture long-range dependencies and improve contextual understanding across sparse oceanic targets. In addition, we compared MambaMPD with mainstream CNN-based segmentation networks. DeepLabV3+ [86] utilises atrous spatial pyramid pooling (ASPP) to encode multi-scale context, while U-Net employs hierarchical skip connections to preserve spatial resolution. Although both achieve reasonable performance, they are constrained by local receptive fields and struggle with long-range semantic modeling. MambaMPD significantly outperforms these baselines, especially in the oil spill category, where it achieves a remarkable 14.41% improvement in mIoU over DeepLabV3+ and a 16.6% gain compared to YOLO-SAM. These results clearly demonstrate the advantage of combining state-space modeling with frequency- and edge-aware modules for structure-sensitive marine pollutant detection. To complement the IoU-based evaluation, Table 6 further reports Precision, Recall, and F1-score. MambaMPD achieves the highest F1-score, confirming its balanced performance across both detection completeness and prediction reliability.

MambaMPD achieves the highest overall mIoU but drops to 43.60% on the Look-alike class, well behind TransOilSeg (62.41%) and SAM-OIL (55.60%). The frequency-enhanced and edge-guided features push the model to detect dark-patch structures

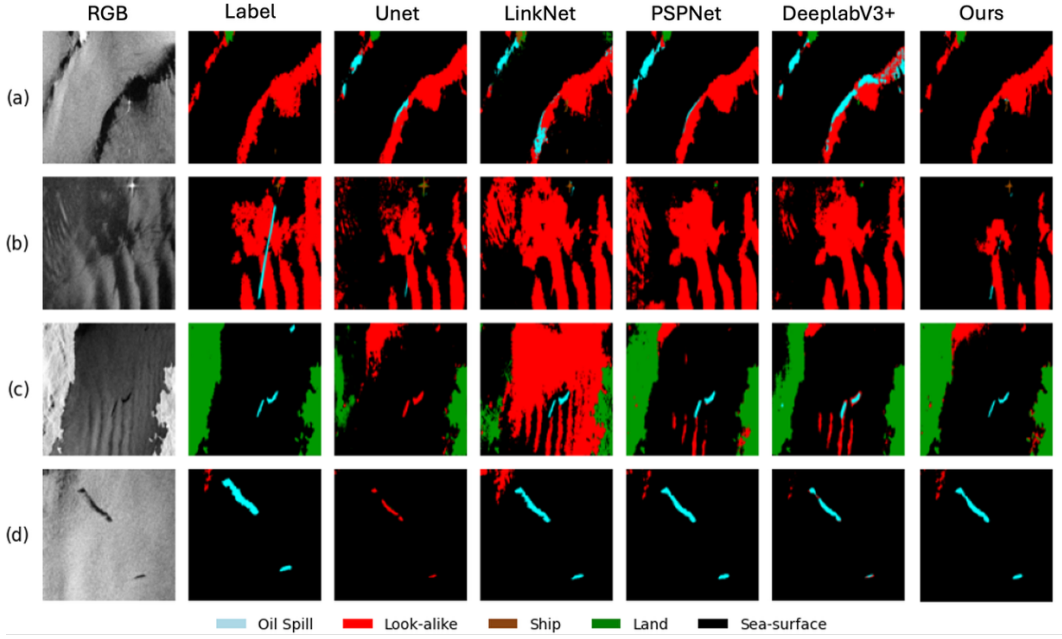


Figure 10. Qualitative performance comparison on the selected test samples of the M4D dataset.

aggressively; this raises Oil Spill IoU to 68.20% (the best among all methods) but also misclassifies some look-alike regions—biogenic films, low-wind areas—as oil spills. TransOilSeg shows the reverse pattern: it leads on Look-alike IoU but falls to 61.38% on Oil Spill. Future work will explore class-aware contrastive learning and look-alike-specific auxiliary losses to better separate these visually similar categories.

4.4.3. Qualitative Analysis

Additional examples on the MADOS test set are shown in Fig. 9. Neither a classical Random Forest classifier nor mainstream encoder-decoder architectures (e.g., U-Net) nor the recent MariNext network succeed in recovering satisfactory edge or texture detail from the multi-spectral data: land-sea boundaries are frequently mis-labelled, and objects with blurred contours are often missed. In contrast, MambaMPD faithfully reconstructs subtle high-frequency structures, such as ship wakes and thin pollutant filaments, while simultaneously suppressing noise. This superior visual performance underscores the model’s capacity to exploit edge-aware priors and long-range context, enabling the reconstruction of details that closely match the true spatial distribution of marine pollutants. A visual comparison with all state-of-the-art baselines is presented in Fig. 10. **In the majority of evaluated scenes**, MambaMPD delivers **more accurate and visually coherent** delineation of pollution targets, **generally outperforming** both traditional convolutional and modern Transformer-based encoders. In rows (a) and (b) of Fig. 10, conventional encoder-decoder networks such as PSPNet misclassify large portions of the slick, reflecting their limited capability to resolve fine-scale boundaries. Although earlier deep learning methods better capture overall shape and edge information, they remain susceptible to texture-induced artefacts. By contrast, MambaMPD obtains sharp, artefact-free borders that follow the ground-truth contour with high fidelity, demonstrating the benefit of its wavelet-enhanced Mamba encoder and edge-guided attention. In the cluttered scene highlighted in Fig. 10, all competing models

Table 5. Ablation study of the proposed modules on the M4D dataset.

FAA	SER	EGA	SeaSurface (% , $\uparrow$)	Oil Spill (% , $\uparrow$)	Look-alike (% , $\uparrow$)	Ship (% , $\uparrow$)	Land (% , $\uparrow$)	mIoU (% , $\uparrow$)	Acc (% , $\uparrow$)
			94.93	61.68	45.14	45.23	91.01	67.51	95.73
✓			95.43	66.97	46.09	37.58	91.82	67.58	96.15
	✓		95.58	66.70	45.51	37.55	93.26	67.72	96.24
		✓	96.40	65.46	46.70	38.36	94.59	68.30	97.00
✓	✓		96.29	65.38	47.62	43.29	94.14	69.34	96.93
	✓	✓	95.73	67.36	47.38	41.88	93.18	69.11	96.49
✓		✓	96.27	67.04	45.16	45.43	95.57	69.48	95.39
✓	✓	✓	96.30	68.20	43.60	49.39	93.79	70.85	97.21

are distracted by background objects and consequently yield false positives. Owing to its stronger discriminative power in visually ambiguous and cluttered scenes, MambaMPD **more reliably** isolates the pollutant region correctly and preserves background integrity. These observations **suggest** that our framework can exploit richer structural priors, thereby improving recognition robustness in **the tested** marine environments.

4.5. Ablation Study

To thoroughly evaluate the effectiveness of each proposed module in our architecture, we conduct extensive ablation experiments on the M4D dataset. **The baseline model is a pure MambaU-Net consisting of a VSS encoder with $\{2, 2, 9, 2\}$ blocks and a standard UnetrUpBlock decoder, without any of the proposed modules (FAA, SER, or EGA). Each module is then individually or jointly added to this Mamba-based baseline to assess its contribution.** The results are shown in Table 5. The investigated modules include: the Frequency-Aware Augmentation Module (FAA) for frequency-aware feature extraction, the SE-ResDecoder (SER) for semantic enrichment, and the Edge-Guided Attention (EGA) for fine boundary preservation. We also analyse the role of their combinations and the cumulative effect when all modules are used together.

4.5.1. Effect of Single Modules

Without applying FAA, SER, or EGA, the baseline MambaU-Net achieves an Oil IoU of 53.90% and an mIoU of 64.97%. This setting serves as the reference point for evaluating the contribution of each individual module.

When only the Frequency-Aware Augmentation (FAA) module is added, the Oil IoU improves significantly from 53.90% to 66.97%, and the mIoU rises from 64.97% to 67.58%. This confirms that FAA enhances the model’s ability to extract discriminative multi-frequency representations, which are crucial for capturing subtle structural, textural, and contextual differences between marine pollutants and ocean backgrounds.

Introducing only the SE-ResDecoder (SER) module yields an Oil IoU of 66.70% and an mIoU of 67.72%. The improvement compared with the baseline demonstrates that SER effectively enriches geometric and structural cues, enabling better discrimination of irregular pollutant shapes within SAR imagery.

Using only the Edge-Guided Attention (EGA) module increases Oil IoU to 65.46% and mIoU to 68.30%, representing the strongest single-module enhancement. This verifies that explicitly modeling boundary priors greatly benefits marine pollution segmentation by mitigating the edge-blurring tendency of SSM-based architectures.

Table 6. Additional evaluation metrics on the M4D dataset. Precision, Recall, and F1-score are computed as the macro-average across all five classes.

Model	Precision (%)	Recall (%)	F1 (%)	mIoU (%)
U-Net	77.80	74.75	75.50	64.97
DeepLabV3+	72.72	77.05	74.57	65.06
TransOilSeg	79.77	78.33	78.99	69.74
OSDMamba	80.13	78.21	77.35	70.25
MambaMPD (ours)	81.02	79.41	78.18	70.85

Table 7. Sensitivity of decoder attention strategies on the M4D dataset. Backbone (VSS {2, 2, 9, 2}), FAA, and SE-ResDecoder remain fixed; only the attention module is replaced.

Attention type	Oil(%)	Ship(%)	Look-alike(%)	mIoU(%)
None (no att.)	68.12	70.35	64.01	69.12
SE	69.44	71.82	65.03	69.88
CBAM	69.87	72.10	65.41	70.12
EGA (ours)	70.58	72.91	66.24	70.83

4.5.2. Effect of Dual Module Combinations

We then evaluate the synergy between two modules. The combination of FAA and SER improves the segmentation accuracy significantly, yielding an oil spill IoU of 65.38% and an mIoU of 69.34%. This suggests that frequency-aware and semantic-enhancing modules complement each other by capturing global context while preserving class-specific features. Combining FAA and EGA (without SER) also achieves strong results, with an oil spill IoU of 64.97% and mIoU of 69.48%, indicating that frequency and boundary information together help the model to suppress noise and highlight transition regions. On the other hand, the SER + EGA setting (without FAA) results in 65.46% oil spill IoU and 68.30% mIoU, which is inferior to the configurations involving FAA, highlighting the foundational role of frequency-aware encoding in marine pollution scenarios.

4.5.3. Full Model with All Modules

Finally, when all three modules (FAA + SER + EGA) are integrated, the model achieves the best performance across nearly all categories. Specifically, it obtains an oil spill IoU of 68.20%, an overall mIoU of 70.85%, and a pixel-level accuracy of 97.21%. This configuration shows superior performance not only in the oil spill class but also across the Sea Surface, Ship, and Look-alike classes, suggesting that each module contributes uniquely and effectively to the final representation. These results clearly demonstrate that the three modules provide complementary advantages: FAA improves global frequency perception, SER enhances semantic discriminability for rare categories, and EGA refines boundaries and edge structures. Their joint effect enables the model to capture both global context and fine-grained details, which are critical for robust oil spill segmentation under real-world, complex marine conditions.

Table 8. Sensitivity of the FAA module to different wavelet bases. Backbone and training protocol remain unchanged; the decomposition level is fixed to $J = 2$. Results are reported on both M4D and MADOS to verify cross-modality robustness.

Wavelet basis	M4D			MADOS		
	OA	mIoU	Oil IoU	OA	mIoU	F1
Haar	97.12	70.83	70.72	82.77	69.16	74.89
Daubechies-2 (db2)	97.05	70.67	70.48	82.75	69.11	74.69
Daubechies-4 (db4)	97.08	70.57	70.61	83.34	69.60	74.71

Table 9. Sensitivity of the FAA module to the wavelet decomposition level J . The wavelet basis is fixed to Haar. Results are reported on both M4D and MADOS.

Level J	M4D			MADOS		
	OA	mIoU	Oil IoU	OA	mIoU	F1
$J = 1$	97.00	70.32	70.05	83.22	69.13	74.15
$J = 2$	97.12	70.83	70.72	83.17	68.93	74.68
$J = 3$	97.06	70.61	70.44	83.47	69.06	74.34

5. Further Analysis

To more comprehensively assess the robustness and efficiency of the proposed MambaMPD framework, we conduct additional analyses from three complementary perspectives: architectural sensitivity to wavelet configurations in the FAA module, decoder attention strategies, and overall computational complexity. These studies provide deeper insights into why MambaMPD exhibits strong boundary modelling capability while maintaining its lightweight nature.

5.1. Sensitivity to Wavelet Configuration in FAA

The FAA module leverages multi-level discrete wavelet transform (DWT) to extract high-frequency components that are critical for modelling thin oil films and fragmented look-alike structures. As wavelet-based frequency decomposition may, in principle, be influenced by the choice of wavelet settings, we evaluate the sensitivity of FAA to two key hyperparameters: (i) the wavelet basis and (ii) the decomposition level J . All experiments are performed on the M4D dataset using the standard VSS backbone configuration $\{2, 2, 9, 2\}$ and the same training protocol as in the main study. As shown in Table 8 and Table 9.

Wavelet basis. We fix the decomposition level to $J = 2$ and examine three commonly used wavelets: Haar, Daubechies-2 (db2), and Daubechies-4 (db4). As summarised in Table 8, the resulting differences in OA, mIoU, and oil-spill IoU are minimal, with mIoU fluctuating within 0.26% and Oil IoU within 0.24%. This demonstrates that the FAA is largely insensitive to the specific wavelet family chosen, confirming that the gains of MambaMPD do not depend on a finely tuned handcrafted basis.

Decomposition level. Next, we vary the decomposition depth $J \in \{1, 2, 3\}$ while fixing the wavelet basis to Haar (Table 9). A shallow single-level decomposition ($J = 1$)

Table 10. Computational complexity comparison. The upper section compares MambaMPD against representative baselines; the lower section isolates the overhead of each proposed module. Flops, latency and FPS are averaged over 100 runs on a single NVIDIA RTX 4070Ti GPU with batch size 1.

Model	Params (M)	FLOPs (G)	Latency (ms)	FPS
U-Net	24.15	58.69	9.12	109.58
DeepLabV3+	57.38	87.48	27.90	55.34
SegNeXt	28.94	31.46	18.05	65.34
OSDMamba	16.14	62.77	12.21	82.42
MambaMPD (ours)	15.23	63.15	11.8	84.7
<i>Module-wise overhead analysis</i>				
Baseline MambaU-Net	15.21	62.84	11.4	87.7
+ FAA	15.21	62.87	11.5	87.0
+ EGA	15.23	63.12	11.7	85.5
+ FAA + EGA (Full)	15.23	63.15	11.8	84.7

results in a slightly reduced mIoU (70.32%), indicating that insufficient frequency resolution weakens multi-scale structure modelling. Increasing the depth to $J = 2$ yields the best performance (70.83%), while $J = 3$ produces a marginal decline due to over-decomposition and loss of fine spatial details. Importantly, the performance remains stable across all settings, highlighting the robustness of FAA to reasonable changes in its wavelet configuration.

5.2. Sensitivity to Attention Strategies

To assess whether the benefits of EGA depend on a particular attention form, we compare EGA against three alternatives while keeping the VSS backbone, FAA, and SE-ResDecoder fixed: (i) no attention, (ii) SE-based channel attention, and (iii) CBAM, a widely used spatial-channel attention module. As shown in Table 7, adding attention generally improves performance over the plain skip-connection baseline. However, EGA consistently achieves the best Oil IoU, Ship IoU, Look-alike IoU, and overall mIoU. The improvements are particularly pronounced in boundary-sensitive categories, demonstrating that explicit integration of multi-scale edge cues provides more effective structural guidance than generic attention mechanisms.

Moreover, the overall variation across attention strategies remains small ($\pm 1.5\%$ mIoU), confirming that MambaMPD is not overly sensitive to the attention formulation. The superior performance of EGA arises from its edge-guided reweighting rather than reliance on a fragile hyperparameter setup.

5.3. Computational Complexity

To validate the claim that the FAA enhances feature representation without increasing computational burden, we quantify model complexity in terms of trainable parameters, FLOPs, and inference latency. FLOPs are computed using `ptflops` under a 512×512 input resolution, and latency is measured as the average over 100 forward passes on an NVIDIA RTX 4070Ti GPU. As reported in Table 10, the FAA introduces no additional trainable parameters and adds only 0.03G FLOPs, translating to a negligible 0.1 ms increase in inference time. EGA likewise incurs only a modest overhead due to its lightweight ECA attention design. The full MambaMPD model increases FLOPs by

merely 0.31G compared with the baseline, while delivering substantial improvements in oil-spill recognition and boundary delineation.

These findings confirm that the enhanced performance of MambaMPD is not a consequence of increased model complexity, but instead stems from the structural integration of frequency-aware and edge-guided modelling. The framework, therefore, achieves a favourable balance between accuracy and efficiency, making it practical for single-GPU deployment. All current experiments use fixed-size patches; scaling to sliding-window inference over full satellite swaths is left to future work.

6. Conclusion

This paper presented MambaMPD, a segmentation framework for marine pollution detection from remote sensing imagery. MambaMPD builds on selective state-space modelling and introduces two modules: Frequency-Aware Augmentation (FAA) and Edge-Guided Attention (EGA), which address the low contrast and boundary ambiguity of marine pollutants in SAR and multispectral images. Mamba offers linear-complexity sequence modelling but is weak at capturing frequency structure and preserving edges; FAA and EGA compensate for these gaps using wavelet transforms and Laplacian-based edge detection, respectively.

On both benchmarks, MambaMPD matches or outperforms existing methods across multiple metrics and its modality-agnostic design allows deployment on both SAR and optical pipelines without architectural changes.

Several limitations should be noted. Look-alike discrimination still lags behind specialised methods such as TransOilSeg, indicating that the learned feature space does not yet separate pollutants from visually similar backgrounds well enough; evaluation is restricted to two benchmarks; broader validation across additional pollution types and sensor modalities is needed, ideally through joint SAR-optical fusion. Future research will focus on extending the framework to multi-modal marine observation scenarios and large-scale monitoring systems, aiming to support more reliable and comprehensive marine pollution assessment.

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